Robotic Manipulation using MoveIt in ROS 2

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I. Introduction

Robotic manipulation requires precise motion planning, collision avoidance, and environment representation. Frameworks such as ROS 2 and MoveIt provide powerful tools for developing such systems.

This work focuses on implementing a structured manipulation pipeline where a robotic arm navigates between predefined container and pan locations. Due to time constraints, the current implementation prioritizes reliability and deterministic execution over full physical realism.

Current Status: The system successfully completes **Task 1** using simplified box-based representations. A demonstration video will be provided with the submission.

II. System Overview

The system architecture consists of three primary components:

A. Robot Model

- 6-DOF manipulator
- Defined using URDF
- Integrated with MoveIt for planning

B. Motion Planning

- Planner: OMPL (RRTConnect)
- Joint-space trajectory generation
- Executed via `move_action`

C. Environment Representation

The workspace is modeled using primitive shapes:

- Table $\rightarrow$ Box
- Containers $\rightarrow$ Grid of boxes
- Pan $\rightarrow$ Box

This significantly reduces computational complexity compared to mesh-based environments.

III. Repository

The implementation is available at:

https://github.com/vedantsinggh/robotic_arm

IV. Current Progress

The current implementation successfully demonstrates Task 1, where the robotic arm navigates between predefined container and pan locations using collision-aware motion planning.

The system operates entirely within RViz and achieves:

- Stable motion planning using MoveIt
- Accurate navigation to predefined targets
- Repeatable task execution

A snapshot of the current system is shown in Fig. 1.

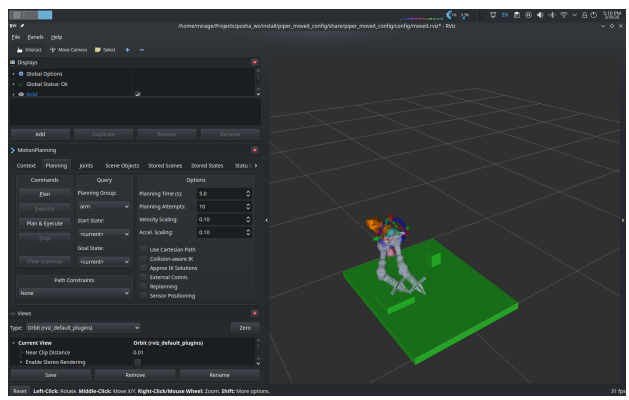

Fig. 1. Current system in RViz showing robotic arm and workspace

A demonstration video of the working system will be provided alongside this report.

V. Workspace Configuration

All objects are defined in the `base_link` frame. The workspace layout is summarized in Table I.

VI. Motion Planning Strategy

A. Joint-Space Planning

Motion is executed using predefined joint configurations. This ensures:

- Stability
- Reduced planning complexity
- Deterministic execution

B. Target Definition

Targets are predefined and mapped to:

- Container positions
- Pan positions
- Home configuration

TABLE I
WORKSPACE OBJECT CONFIGURATION

Object	X (m)	Y (m)	Z (m)
Table	0.50	0.00	-0.025
Container Row 1	0.30 – 0.50	-0.10	0.05
Container Row 2	0.30 – 0.50	0.10	0.05
Pan	0.50	-0.25	0.025

VII. INVERSE KINEMATICS FORMULATION

Let the joint configuration be:

$$\mathbf{q} = [q_1, q_2, \ldots, q_n]^T \tag{1}$$

The forward kinematics is given by:

$$\mathbf{x} = f(\mathbf{q}) \tag{2}$$

The inverse kinematics problem is:

$$\mathbf{q} = f^{-1}(\mathbf{x}) \tag{3}$$

MoveIt solves IK numerically using iterative methods based on the Jacobian:

$$\mathbf{q}_{k+1} = \mathbf{q} * k + J^{\dagger}\left(\mathbf{x} * target - f(\mathbf{q}_k)\right) \tag{4}$$

where $J^{\dagger}$ is the pseudo-inverse of the Jacobian matrix.

VIII. TASK EXECUTION PIPELINE

The manipulation task is defined as:

1) Move to container location
2) Move to pan location
3) Return to home

A menu-driven interface allows both manual and automated execution.

IX. COLLISION HANDLING

Collision avoidance is managed using the MoveIt planning scene:

- Objects are added as collision primitives
- The planner ensures valid trajectories

X. USE OF AI IN DEVELOPMENT

Artificial Intelligence tools were utilized during the development process primarily for:

- Understanding ROS 2 and MoveIt architecture
- Debugging errors and interpreting system logs
- Exploring different approaches for motion planning and environment modeling

However, the final implementation was not directly generated or blindly adopted. Instead, all solutions were applied only after thorough study and understanding of the underlying robotics concepts and framework behavior.

This approach ensured:

- Correct integration with ROS 2 ecosystem
- Reliable motion planning pipeline
- Avoidance of unstable or non-deterministic solutions

Thus, AI served as a *support tool for research and guidance*, while the actual system design and implementation decisions were made through deliberate engineering reasoning.

XI. RESULTS

The system demonstrates:

- Collision-aware planning
- Reliable motion execution
- Repeatable task performance

XII. LIMITATIONS

- No gripper or grasp execution
- No perception system
- No physics simulation
- Static target positions

XIII. FUTURE WORK

Future improvements include:

- Integration with Gazebo for physics simulation
- Gripper and grasp planning
- Vision-based object detection
- Dynamic task planning

XIV. CONCLUSION

This work presents a robust robotic manipulation pipeline using MoveIt in ROS 2. By simplifying the environment and focusing on reliable planning, the system achieves stable task execution. The current implementation provides a strong foundation for future extensions into full manipulation systems.